DATA SCIENCE & ANALYTICS

MODULE 04 SESSION 17

Power Query II: files that arrive as files

Append, a folder of files, the store list and Merge, across two lectures.

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Sun 1 Nov 2026

ACROSS THE FILE-PREPARATION LESSONS

The Power Query sequence

01 Append queries, and say why names matter

02 Combine a folder of files into one table

03 Split columns and resolve error cells

04 Build a clean query as a Reference

05 Merge with the right join kind

EVERYONE DO

Step 1.1

Review Modules 1–3

Nakheel data, the SQL you wrote, and choosing a model for a business question. Then your first Power BI report.

INSTRUCTOR DO

Step 1.2

How much did the shops sell in each governorate?

Two yearly files from finance, twelve files from the shops, a store list typed by hand.

`power-query/labs/D17/README.md · download, extract, check`

YOU KNOW THESE FROM SQL · TODAY IN POWER QUERY

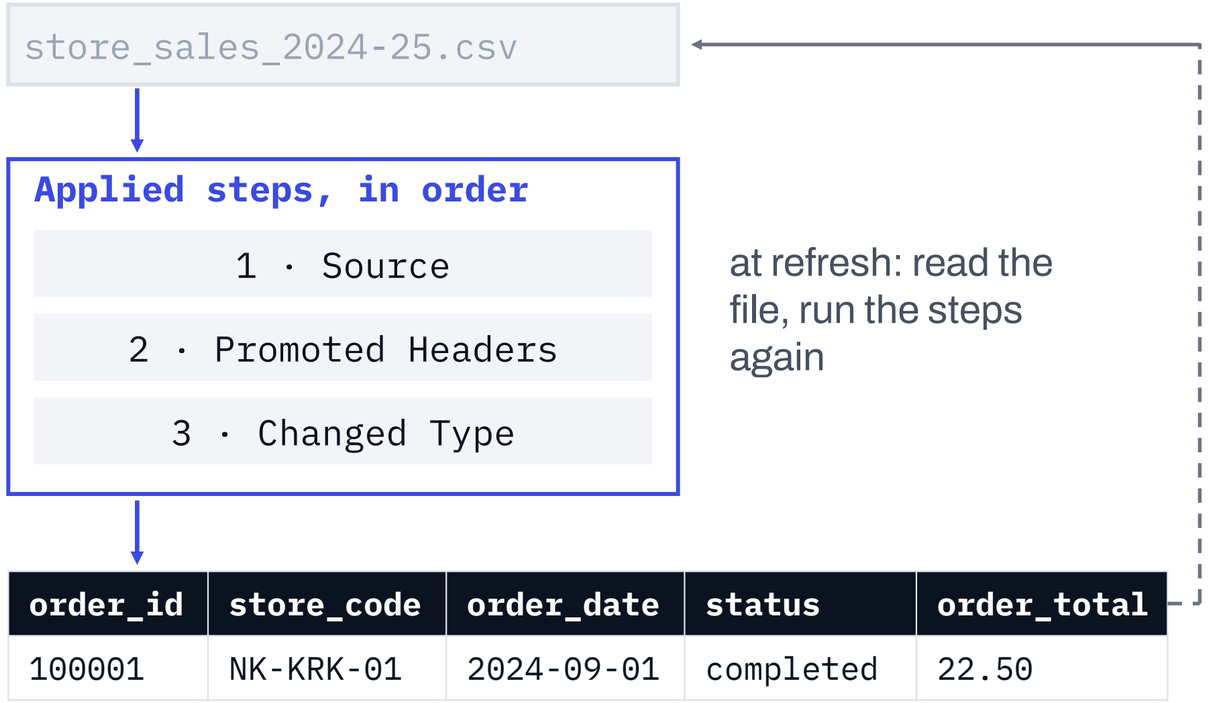
The same ideas, a new place

SQL you wrote	Power Query today	Where
UNION ALL	Append queries	activity 01
many tables stacked	a folder of files, Source.Name	activity 02
SPLIT, SAFE_CAST	Split Column, Replace Errors	activities 04–05
a view on a table	a Reference query	activity 04
LEFT JOIN, and … IS NULL	Merge: Left Outer, Left Anti	activities 06–07

Nakheel SQL check: completed store orders total 205,338.00 dinars.

A query is a table and the steps that make it

A query is one table and the steps that make it, listed by name in the Queries pane. **Applied steps** lists them; they run in order at every refresh.



PQ · 01

Two years in one table

Finance sends store sales once a year. Stack the two files, and find out why half the revenue is missing.

1 · store_sales	2 · store_files_sales	3 · stores	4 · store_sales_by_store
Append	Folder	Reference, Split	Merge

Append puts the rows of one query under the rows of another, in one table: the Power Query version of UNION ALL. Choose Append queries as new, so the two inputs stay as they are.

INSTRUCTOR DO

Step 2.1

Append is UNION ALL, matched by name

Two yearly files, one renamed column, and a card that reads less than half.

power-query/labs/D17/01-Evr-Append-Two-Years/README.md

BEFORE → APPEND QUERIES AS NEW → AFTER · THE TWO FINANCE FILES

Append lines columns up by name

store_sales_2024-25.csv · 892 rows

order_id	store_code	order_date	status	order_total
100001	NK-KRK-01	2024-09-01	completed	22.50

store_sales_2025-26.csv · 1,009 rows

order_id	order_date	store_code	status	amount
101340	2025-09-01	NK-AMM-03	completed	30.00

indigo: the same names, in another order · **amber**: a new name

store_sales · 1,901 rows · 6 columns

order_id	store_code	order_date	status	order_total	amount
100001	NK-KRK-01	2024-09-01	completed	22.50	*null*
100003	NK-AMM-01	2024-09-03	completed	47.00	*null*
101340	NK-AMM-03	2025-09-01	completed	*null*	30.00
101341	NK-AMM-03	2025-09-01	cancelled	*null*	25.00

In SQL, UNION ALL pairs by position: the date **2025-09-01** would land under store_code, or the types would clash.

By name: the swap did no harm; the new name became a sixth column (amber).

WHAT COMES BACK · A CARD ON SUM OF order_total · COMPLETED STORE ORDERS

Half the revenue sits in a column the card does not add

Completed store revenue in SQL: **205,338.00**

order_total, 2024-25	97,743.00	amount, 2025-26	107,595.00

`Append as it came`	the card says 97,743.00: only the first year	×
`Combine the amount columns later`	a saved step also replays, but leaves the inputs with different names	✓
`Rename amount at the source`	the card says 205,338.00, at every refresh	✓

Rename amount in store_sales_2025-26: every dependent query then reads one order_total column.

EVERYONE DO

Step 2.2

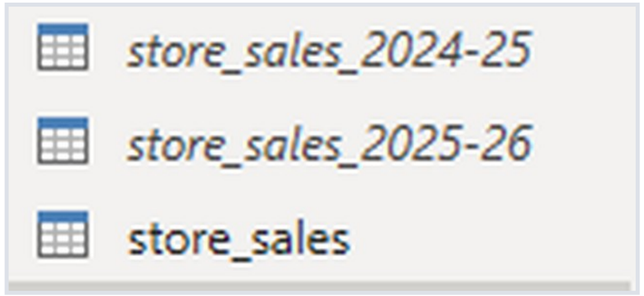
Append two years, and keep the inputs out of the model

Activity 01: append, find the missing revenue, fix it at the source.

power-query/labs/D17/01-Evr-Append-Two-Years/README.md

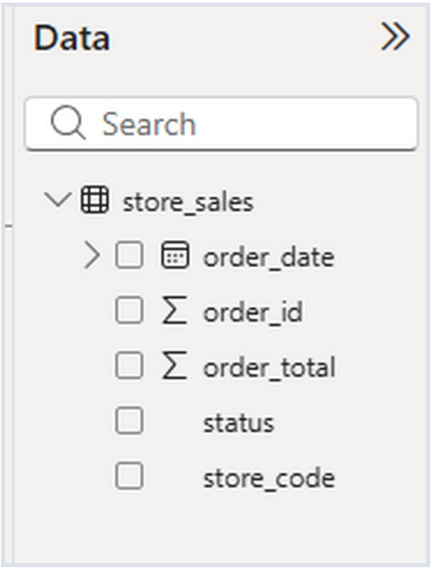
Italic queries feed others; only the result is loaded

Power Query · the Queries pane



italic: Enable load is off. These queries still run at every refresh, because store_sales reads them.

the report · the Data pane



only store_sales: 1,901 rows, 5 columns

The yearly pieces stay out of the model; only their result, store_sales, is loaded. Screens: Power BI Desktop, November 2026.

THE RULE TO KEEP

Fix the input, not the result.

A fix in the query that reads the file reaches every query built on it, at every refresh.

PQ · 02

Combine a folder of shop files into one table

Each shop sends one file named after itself. Read the folder, not the files.

1 · store_sales	**2 · store_files_sales**	3 · stores	4 · store_sales_by_store
Append	Folder	Reference, Split	Merge

INSTRUCTOR DO

Step 3.1

Read a folder of CSV files with Combine Files

Look at the file list, keep the CSV files, combine, and turn the file name into a store code.

power-query/labs/D17/02-Evr-Combine-A-Folder/README.md

Combine Files runs the same steps on every file in the folder

one row per file

Content	Name
Binary	NK-AMM-01.csv
Binary	NK-AMM-02.csv
Binary	NK-AMM-03.csv
Binary	NK-AQB-01.csv
Binary	NK-IRB-01.csv
Binary	NK-IRB-02.csv
Binary	NK-KRK-01.csv
Binary	NK-MDB-01.csv
Binary	NK-MFQ-01.csv
Binary	NK-POP-01.csv
Binary	NK-SLT-01.csv
Binary	NK-ZRQ-01.csv
Binary	**notes.txt**

each →

the recipe, written once

Source: one file

Promoted Headers

Changed Type

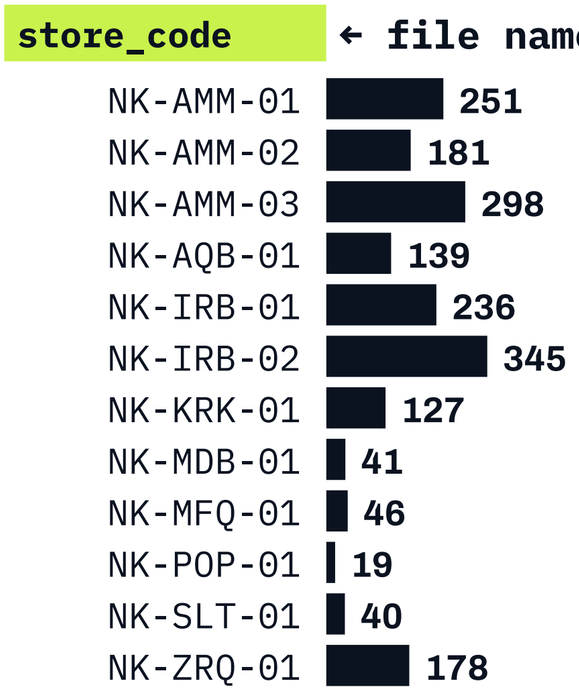
run on each of the 12 CSV files

Power Query keeps it under **Helper queries** as Transform Sample File.

notes.txt is not sales: filter it out first, or the recipe reads it wrongly.

→

store_files_sales · 1,901 rows



rows per file; total **1,901**

A 13th shop sends a file at refresh its rows join the table: no new query ✓

EVERYONE DO

Step 3.2

Activity 02: combine the twelve shop files into one table

Activity 02: connect to the folder, keep the CSV files, combine, name the store.

power-query/labs/D17/02-Evr-Combine-A-Folder/README.md

STUDENT DO

Activity 03: revenue by store and by month

Three questions a regional manager would ask, from the table you just combined.

power-query/labs/D17/03-Stu-Revenue-By-Store-And-Month/README.md

STUDENT DO

Questions on revenue by store and by month

INSTRUCTIONS

From store_files_sales, completed orders: 1. Which shop sold the most? 2. Which calendar month was best, and which worst? 3. Which store code is not a shop?

Full steps and Check yourself: power-query/labs/D17/03-Stu-Revenue-By-Store-And-Month/README.md

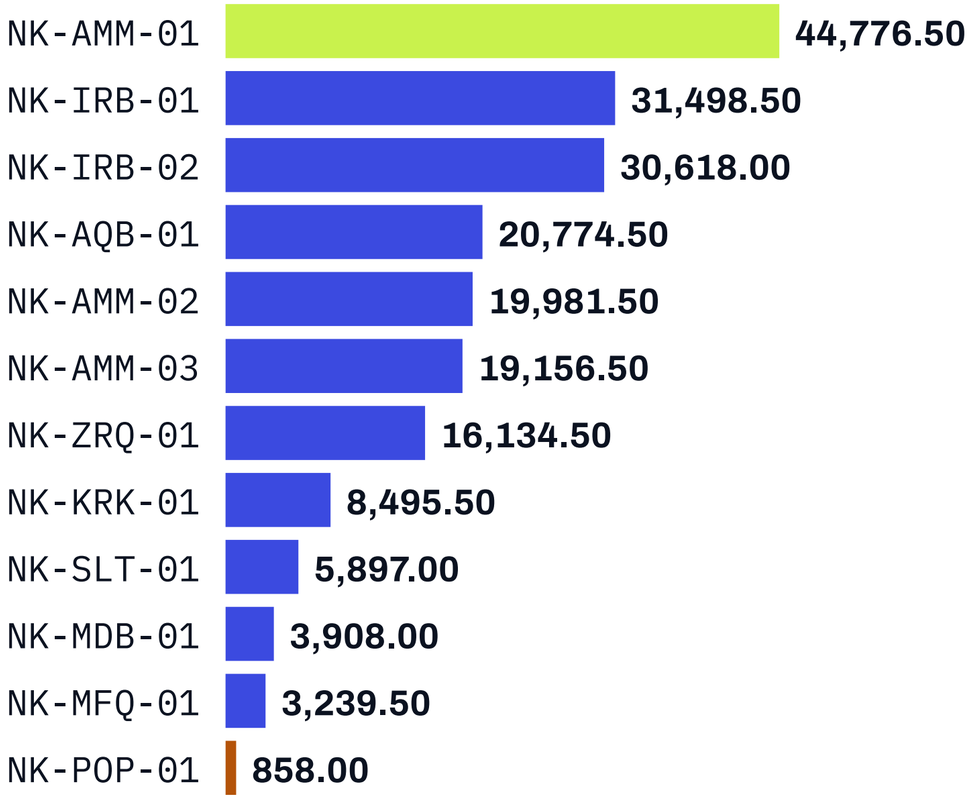
REVIEW

Time's up.
Let's review.

Which shop, which month?

`Solved: power-query/labs/D17/03-Stu-Revenue-By-Store-And-Month/Solved/solution.md`

Revenue by store: Abdali leads and NK-POP-01 is a stall





The twelve bars add up to 205,338.00. NK-POP-01 (amber) is the Ramadan stall: 19 orders, 18 of them completed, 15–28 March 2025. After the break it finds no store.

THE RULE TO KEEP

Add a file to the folder and it is included at the next refresh.

Add a file with the same columns and new orders, then refresh. Overlapping exports would count the same orders twice.

PQ · 03

The store list needs cleaning before we can use it

Three facts in one column, sizes as text, one size unknown. Clean it in a query that reads the raw one.

1 · store_sales	2 · store_files_sales	**3 · stores**	4 · store_sales_by_store
Append	Folder	Reference, Split	Merge

A Reference is a new query that starts from another query’s result. Change the first, and the reference follows; a Duplicate copies the steps and then changes independently.

INSTRUCTOR DO

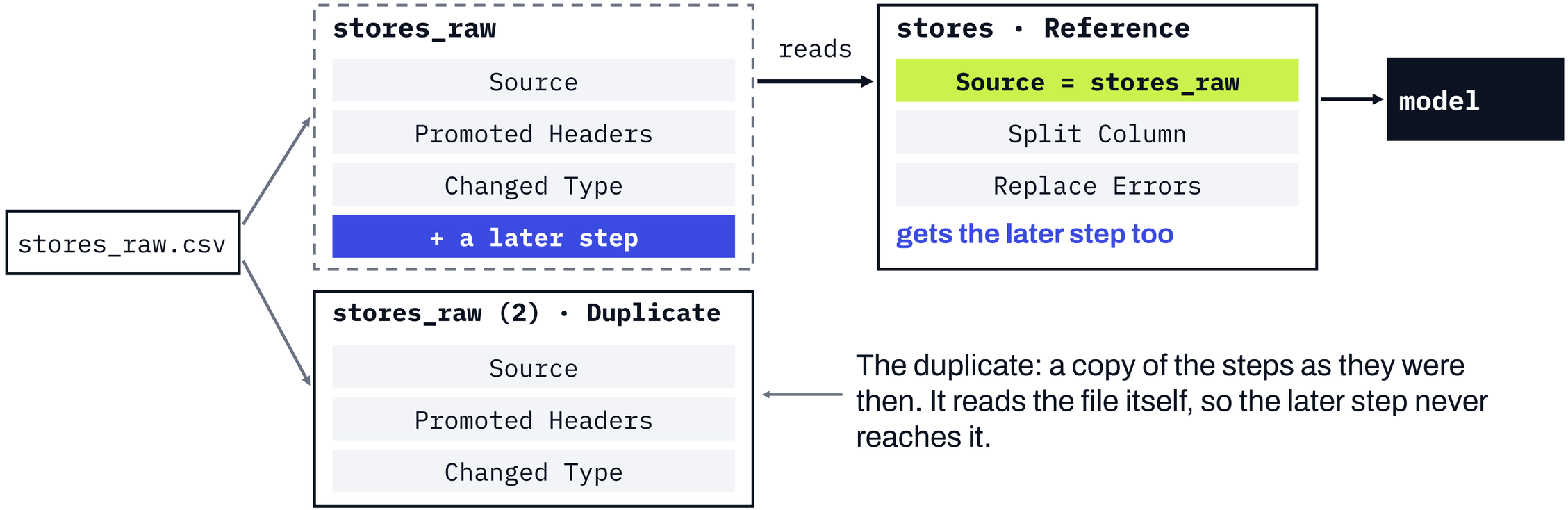
Step 4.1

Clean the store list in a Reference query and keep the raw file unchanged

Keep stores_raw exactly as the file; clean in stores, a reference to it.

power-query/labs/D17/04-Evr-Split-Reference-Errors/README.md

Reference and Duplicate: what each one creates



Dashed: Enable load off. stores_raw still runs at refresh, because stores reads it; only stores goes into the model.

A copy is right	you need it as it was: lab 09 keeps its orders, one row each	✓

EVERYONE DO

Step 4.2

Activity 04: split the city column and handle Jerash’s unknown size

Activity 04: three facts into three columns, a size into a number, Jerash’s unknown size.

power-query/labs/D17/04-Evr-Split-Reference-Errors/README.md

SPLIT COLUMN · BY DELIMITER · WHERE IT CUTS

Split Column by delimiter: the three options

stores_raw · governorate - city - area

AM	␣-␣	Amman	␣-␣	Abdali

␣ marks a space: the delimiter is space, hyphen, space

Each occurrence	AM	Amman	Abdali		3 columns: code, governorate, area
Left-most	AM	Amman - Abdali			2 columns: the first cut only
Right-most	AM - Amman	Abdali			2 columns: the last cut only
size	420	m²	unknown	null	the first space: number, unit

The delimiter "-" without its spaces

AM␣	␣Amman␣	␣Abdali

Each value keeps a space (␣). A filter on Amman finds 0 of its 3 shops.

WHAT COMES BACK · SPLIT AT EACH “ - ”

Split the city column into store code, governorate and area

governorate - city - area	gov_code	governorate	area
AM - Amman - Abdali	AM	Amman	Abdali
IR - Irbid - Irbid	IR	Irbid	Irbid
BA - Balqa - Salt	BA	Balqa	Salt
JA - Jerash - Jerash	JA	Jerash	Jerash

The header says city; the values are a code, the governorate and the area. The governorate column has 9 values.

TWO KINDS OF ERROR · THE STEP-LEVEL MESSAGE IS ILLUSTRATIVE

Two kinds of error: one bad cell, or a step that stops the query

cell-level · size.1 as Whole Number

size (text)	size.1
420 m²	420
380 sqm	380
260 m2	260
unknown (Jerash)	**Error**

The table still shows; one cell holds a message about one value. Fix it before loading.

step-level · a step fails

```
Expression.Error: The column 'store_name' of
the table wasn't found.
```

no table to preview

Nothing shows; a step must change. Here a later step still uses store_name after a rename upstream.

A Reference inherits a later step, including one that breaks a step after it.

QUICK CHECK

Jerash’s size is unknown. Remove Errors or Replace Errors?

Select size.1. Remove Errors deletes rows with an error there. Replace Errors puts a value of your choice, such as null, in the cell and keeps the row (SQL’s SAFE_CAST gives NULL). Which one keeps the store list true?

KEEP ERRORS FIRST, TO SEE THE ROWS · THEN DECIDE BY WHAT THE VALUE MEANS

Choose how to fix an Error cell by what the value means

	Remove Errors	Replace Errors → null	Fix the text first
Jerash: **unknown** the size is not known	× 11 shops: Jerash leaves the list; no join can find it	✓ 12 shops; size empty; 2,880 m² known	– nothing to fix: no number in the text
Aqaba till, sale 102869: **93.50 JOD** a number typed wrong	× completed 1,939.00: the 93.50 sale is gone	× completed 1,939.00: row kept, 93.50 lost	✓ delete the text JOD, then set the type: 2,032.50, the total head office expects

Replace Errors is right when the value is not known. When the number is there, repair the text before the type. **EXAMPLE:** lab 10's till file.

STUDENT DO

Activity 05: clean the rest of the store list

Latitude and longitude, a short name for charts, and the facilities manager’s question.

power-query/labs/D17/05-Stu-Clean-The-Store-List/README.md

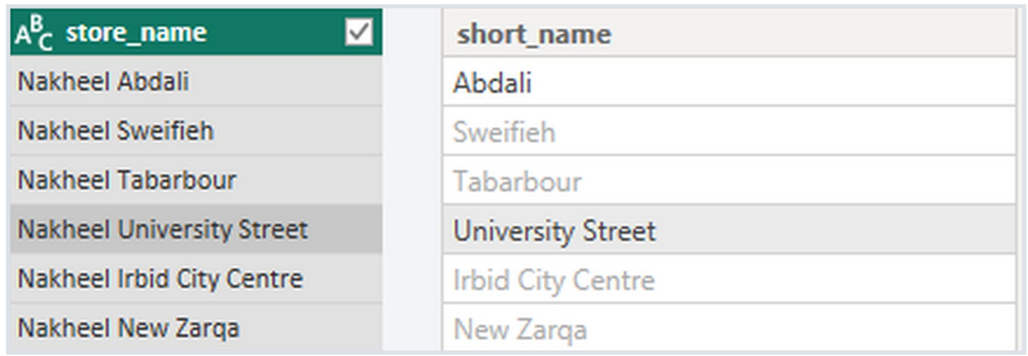
Column From Examples and the rule it writes

Column From Examples makes a column from values you type; Power Query writes one step that fills every row, and shows its formula.

A: after the first space · B: the last word

store_name	rule A	rule B
Nakheel Abdali	**Abdali**	**Abdali**
Nakheel Sweifieh	Sweifieh	Sweifieh
Nakheel Tabarbour	Tabarbour	Tabarbour
Nakheel University Street	**University Street**	**Street**
Nakheel Irbid City Centre	Irbid City Centre	**Centre**
Nakheel New Zarqa	New Zarqa	**Zarqa**

Desktop: typed in black, filled in grey



A second example, University Street, fits only rule A.

DATA TYPES · THE SAME TEXT, TWO LOCALES

The locale decides how Change Type reads 30.00 and 31.9632

	text in the file	English (United States)	a decimal-comma locale
latitude	31.9632	**31.9632**	**319632**
order_total	30.00	**30.00**	**3000**
the completed card		**205,338.00**	**20,533,800.00**

In English (United States) the dot is the decimal point; with a decimal comma it separates thousands. **Using Locale** sets the type by the file's locale.

STUDENT DO

Questions on the position, short name and floor area

INSTRUCTIONS

In stores: 1. Split the position into latitude and longitude. 2. Add short_name from examples: Abdali, University Street. 3. Is opened a date? 4. How much selling floor in Amman, and in the whole country?

Full steps and Check yourself: power-query/labs/D17/05-Stu-Clean-The-Store-List/README.md

REVIEW

Time's up. Let's review.

The store list

Solved: power-query/labs/D17/05-Stu-Clean-The-Store-List/Solved/solution.md

THE ANSWERS · STORES, 12 ROWS

The cleaned store list, with coordinates and floor area

store_code	short_name	governorate	latitude	longitude	size_m2
NK-AMM-01	Abdali	Amman	31.9632	35.9106	420
NK-IRB-01	University Street	Irbid	32.539	35.856	310
nk-irb-02	Irbid City Centre	Irbid	32.5556	35.85	240
NK-JRS-01	Jerash	Jerash	32.2808	35.8993	null

Floor area: Amman 1,060 m² · all shops 2,880 m² (Jerash unknown). Look again at row 3’s code.

Break

Stretch, then Merge

After the break: the sales meet the store list.

PQ · 04

Merge the sales with the store list

Only the store list knows the governorate. Join it to the sales, and read the match count first.

1 · store_sales	2 · store_files_sales	3 · stores	**4 · store_sales_by_store**
Append	Folder	Reference, Split	Merge

Merge finds, for each row of the first query, the rows of the second with the same key, and attaches them as a table you expand: the Power Query version of a JOIN.

INSTRUCTOR DO

Step 5.1

Merge store_sales with stores and read the match count

store_sales on the left, stores on the right, by store_code, Left Outer.

power-query/labs/D17/06-Evr-Merge-Sales-And-Stores/README.md

MERGE · JOIN KIND · FIVE STORE SALES AND FOUR STORES, AFTER UPPERCASE

The six join kinds in Merge and the rows each one returns

Sales left, stores right, no NK-. **Ink**: matched, kept. **Amber**: no partner, kept, other side empty. **Grey**: not returned.

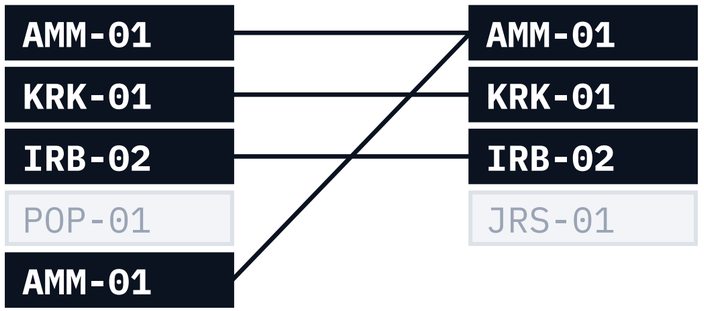
Left Outer LEFT JOIN

5 rows · full data 1,901



Inner JOIN

4 rows · full data 1,882



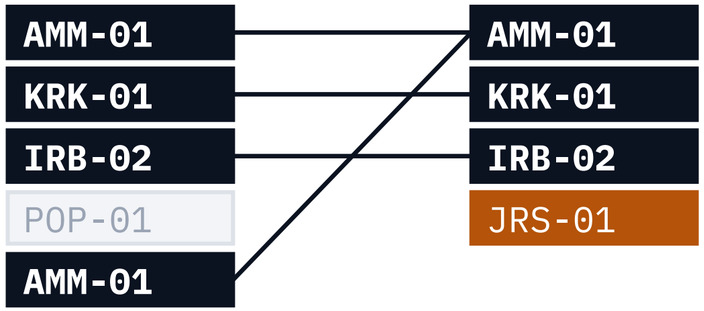
Full Outer FULL OUTER JOIN

6 rows · full data 1,902



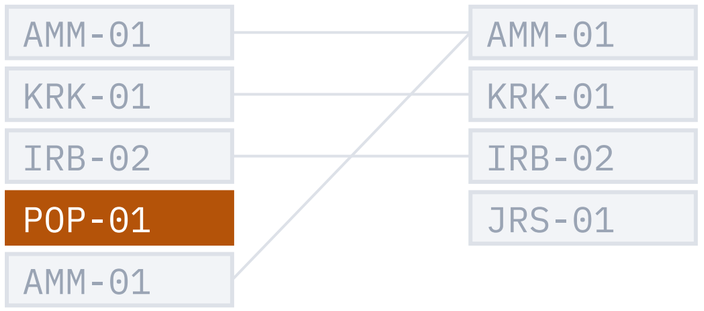
Right Outer RIGHT JOIN

5 rows · full data 1,883



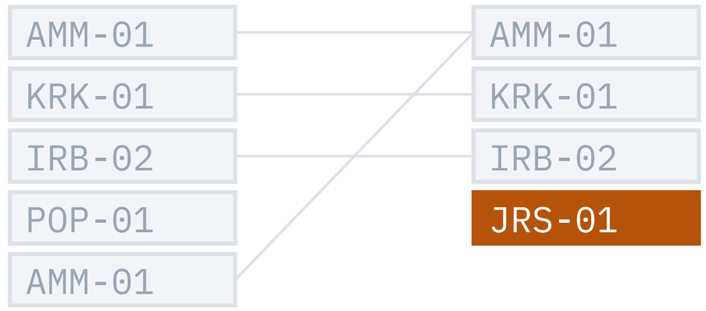
Left Anti LEFT JOIN … IS NULL

1 row · full data 19



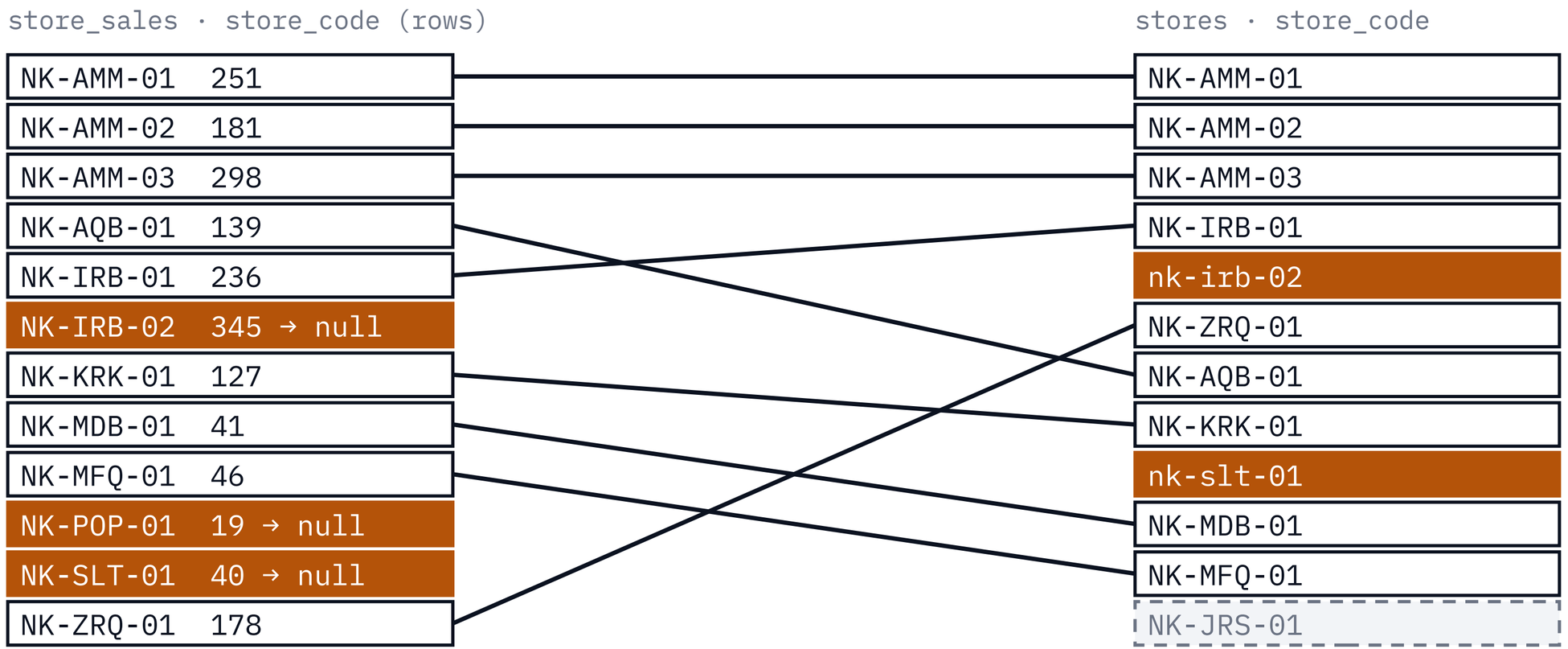
Right Anti RIGHT JOIN … IS NULL

1 row · full data 1



MERGE · LEFT OUTER · EXPECTED FULL-DATA COUNTS, BEFORE THE KEY FIX

Left Outer merge: 404 sales from three codes find no store



1,497
of 1,901 match

Amber: no partner. Power Query compares text exactly: items that differ only in case are “different, not identical” (Microsoft). Dashed: Jerash, no sales yet.

AFTER OK · THE NEW COLUMN HOLDS A TABLE IN EVERY ROW

Expand the store’s columns you need

store_sales_by_store · the merge adds a column called stores

order_id	store_code	order_total	stores
100003	NK-AMM-01	47.00	**Table**
100001	NK-KRK-01	22.50	**Table**

Each Table is that sale’s matching rows of the store list, one row when the key matches, none when it does not.



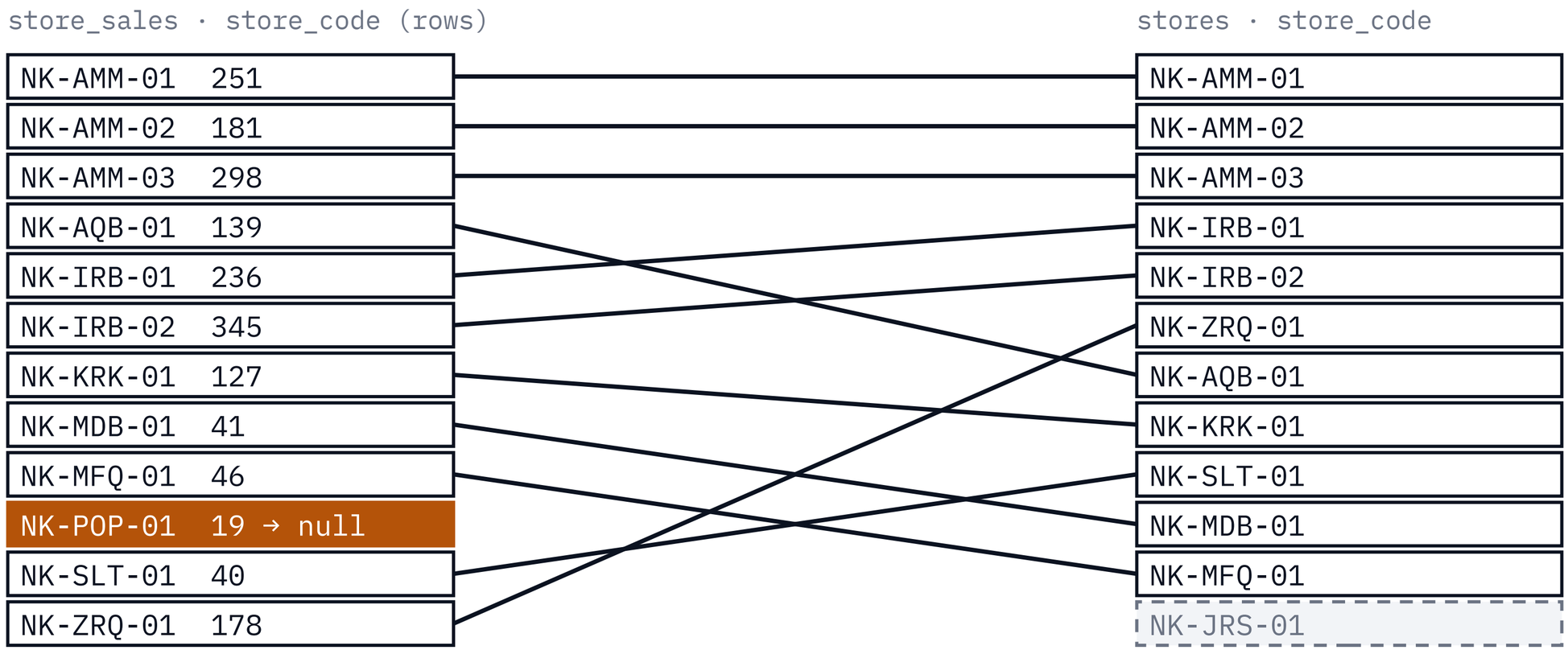
expand governorate, area, short_name · untick “Use original column name as prefix”

order_id	store_code	order_total	governorate	area	short_name
100003	NK-AMM-01	47.00	**Amman**	**Abdali**	**Abdali**
100001	NK-KRK-01	22.50	**Karak**	**Karak**	**Karak**

Merge adds one column of tables; expanding copies the store’s columns into every sale. The 404 with no store keep empty columns.

MERGE · LEFT OUTER · AFTER UPPERCASE ON store_code IN stores

After the uppercase fix, the stall and Jerash have no match

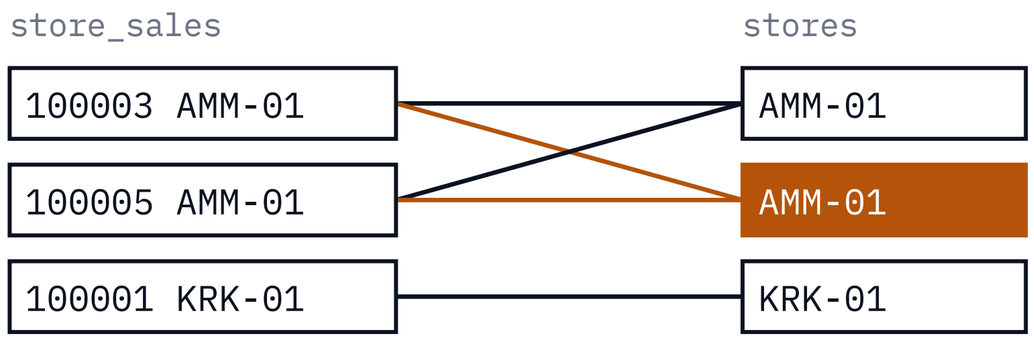


1,882
of 1,901 match

One fix in stores; the merge reads stores again at refresh. 19 sales stay unmatched: NK-POP-01, the stall.

ILLUSTRATIVE · IF THE STORE LIST HAD NK-AMM-01 TWICE

A store code listed twice in stores adds extra sales rows



store_sales_by_store · 5 rows, not 3

100003 AMM-01
100003 AMM-01
100005 AMM-01
100005 AMM-01
100001 KRK-01

each Abdali sale finds two store rows, so it comes back twice

Rows after the merge	2,152, not 1,901	✗
Completed revenue	250,114.50, not 205,338.00	✗
The check	Keep Duplicates on store_code finds a repeated key	✓

The match message does not warn you; the row count does. D06 said the same of SQL joins.

EVERYONE DO

Step 5.2

Activity 06: merge, fix the store code and chart revenue by governorate

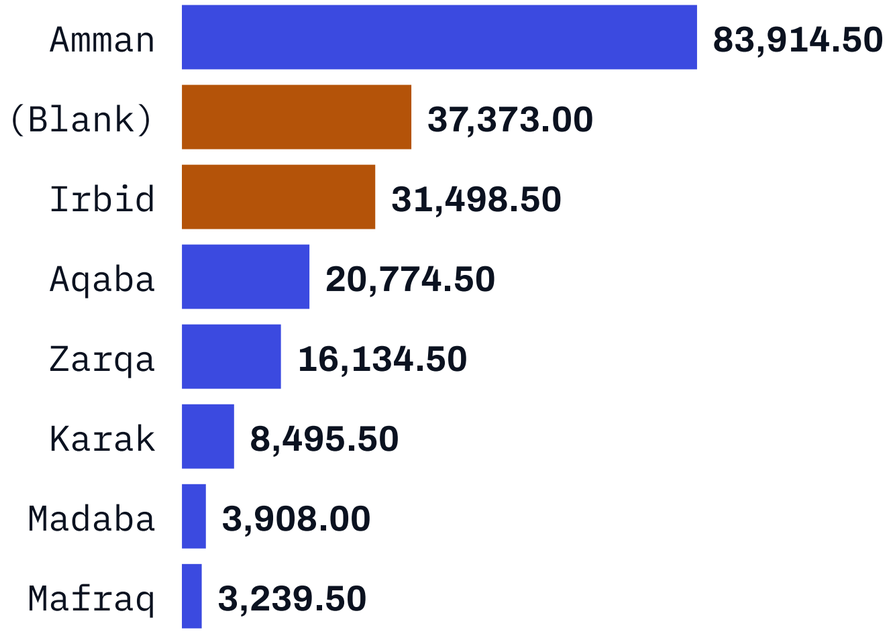
Activity 06: merge, expand, find the 404, UPPERCASE in stores, then a bar chart.

power-query/labs/D17/06-Evr-Merge-Sales-And-Stores/README.md

WHAT COMES BACK · COMPLETED REVENUE BY GOVERNORATE · LEFT OUTER

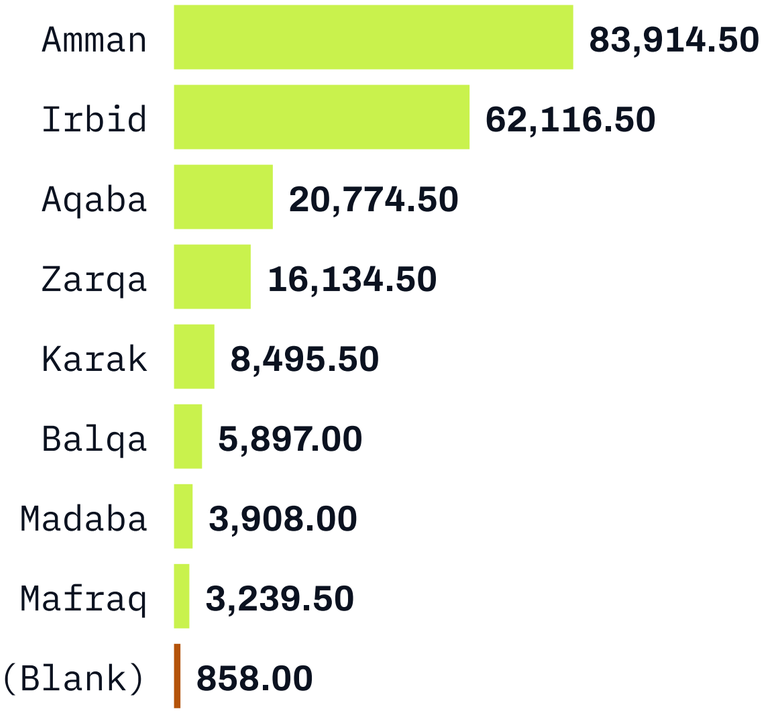
Before the key fix, 36,515.00 of revenue has no governorate

BEFORE · the codes as typed



amber: wrong · Irbid lacks IRB-02 · no Balqa bar

AFTER · UPPERCASE in stores



Both charts: store_sales_by_store, completed orders, total 205,338.00. (Blank) keeps the stall's 858.00.

STUDENT DO

Activity 07: sales without a store and a store without sales

An auditor’s three questions after any merge: Left Anti, Right Anti and Inner.

power-query/labs/D17/07-Stu-Sales-Without-A-Store/README.md

STUDENT DO

Find the sales with no store and the store with no sales

INSTRUCTIONS

Start each merge from store_sales. 1. Which sales have no store? 2. Which store has no sales? 3. What would an Inner merge leave out of revenue by governorate?

Full steps and Check yourself: power-query/labs/D17/07-Stu-Sales-Without-A-Store/README.md

REVIEW

Time's up.
Let's review.

Who has no match?

`Solved: power-query/labs/D17/07-Stu-Sales-Without-A-Store/Solved/solution.md`

The join kind that answers each question about a merge

matched to a store 204,480.00

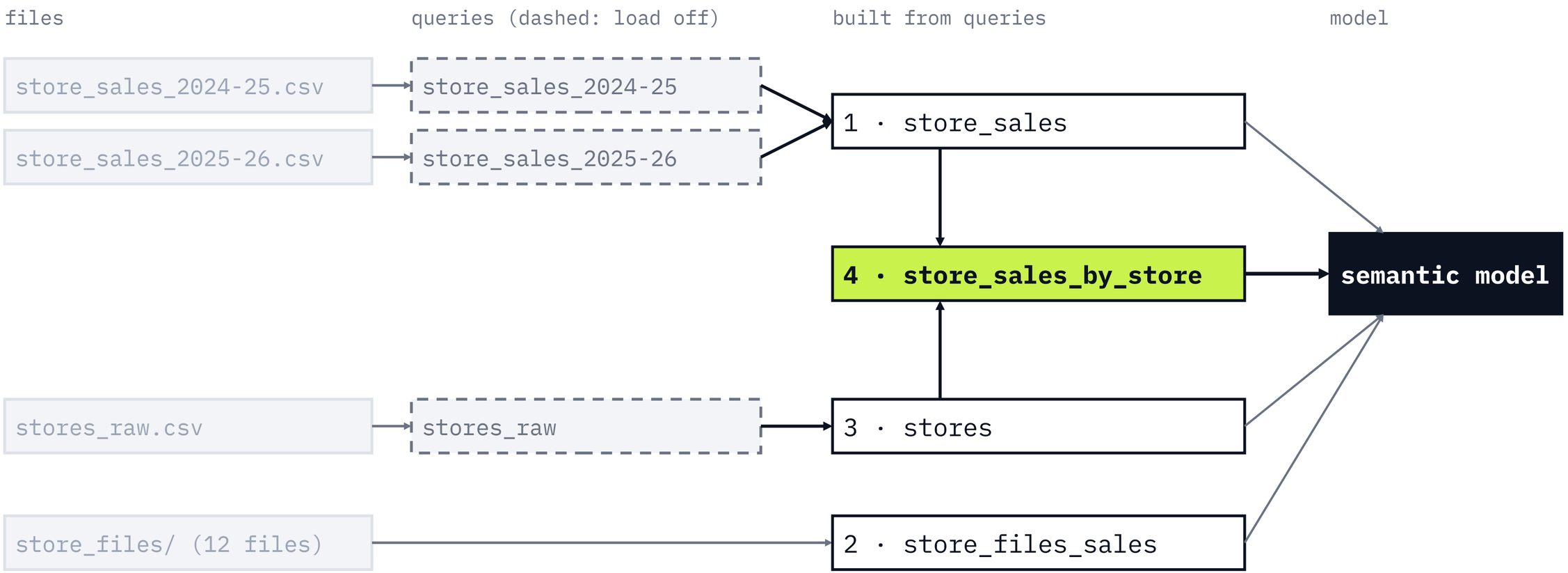
NK-POP-01, 18 completed sales: 858.00

Inner	revenue by governorate? 204,480.00: 858.00 short of the books	✗
Left Outer	revenue by governorate? 205,338.00, with a (Blank) bar to ask about	✓
Left Anti	sales with no store? 19 rows, all NK-POP-01 (18 completed)	✓
Right Anti	a shop with no sales? NK-JRS-01, Jerash: opens after the data ends	✓

For revenue by governorate, use Left Outer and say what the blank is.

THE QUERIES YOU BUILT TODAY · NUMBERED BY PART

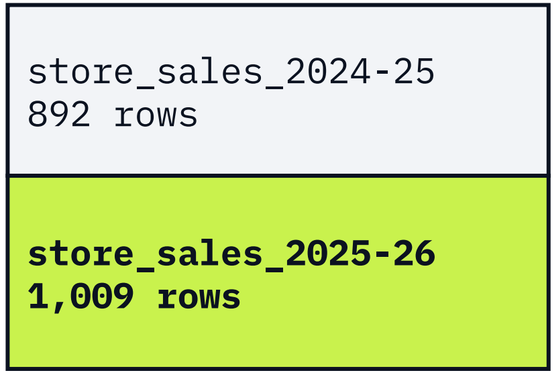
The queries we built today and what each one loads



Dashed: load off. store_sales and store_files_sales hold the same orders by two routes, kept today to compare: each totals 205,338.00, never add them.

How Append, Merge and a relationship combine two tables

Append (today)



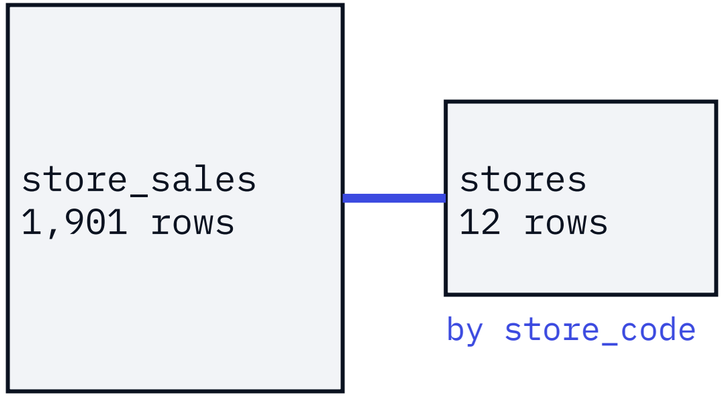
1,901 rows, the same 5 columns

Merge (today)



1,901 rows, 3 more columns; Amman written on 730 rows

Relationship (D18)



Both tables stay as they are; the model follows the line

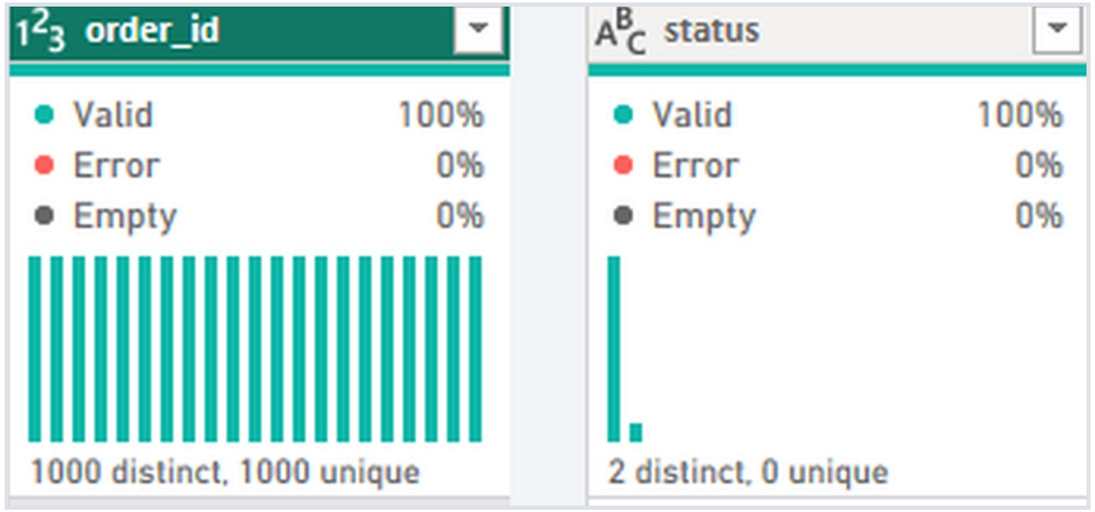
Revenue by governorate, completed orders:

Append	more rows, still no governorate column: not this question	✗
Merge	Amman 83,914.50 … (Blank, the stall) 858.00: total 205,338.00	✓
Relationship	the same bars on D18, and stores keeps its 12 rows	✓

COLUMN PROFILE · TOP 1,000 ROWS · STORE_SALES AFTER THE RENAME

The column profile counts only the first 1,000 rows by default

Desktop: order_id and status, top 1,000 rows

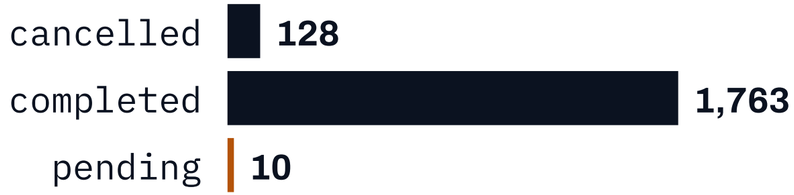


`1000 distinct` order ids and **`2 distinct`** statuses, on a table of 1,901 orders

top 1,000 rows



entire data set · 1,901 rows



pending: 10 orders, the first at row 1,887

Power Query reads samples elsewhere too: Combine Files sets types from the first 200 rows, and the merge dialog shows an estimate. Check the whole table.

THE RULE TO KEEP

The match count in the Merge dialog is an estimate, so check the merged table.

After expanding, check the row count, unmatched rows and completed total. A matching total alone can hide the wrong groups.

STUDY LIST · IN CLASS AND IN THE LABS

The Power Query features used today

In class	What it does	Activity
Append queries as new	rows of two queries in one table (UNION ALL)	01
Enable load	a helper query runs but is not loaded	01, 04
Folder → Combine Files	every file in a folder, with Source.Name	02
Reference	a query that starts from another query	04
Split Column → By Delimiter	one text column into several	04, 05
Keep / Replace Errors	find error cells; replace an unknown value with null	04
Column From Examples	a column from values you type	05
Merge queries as new	a join: Left Outer, Inner, the anti joins	06, 07

Lab	After class, self-guided, not submitted
08	**Desktop local paths: Manage Parameters; a status parameter**
09	**JSON → To Table, expand records and lists; one row per item**
10	**Remove Top Rows, Use First Row as Headers, errors that are sales**
11	**Unpivot Other Columns, Group By, a merge on two keys (optional)**

CLOSE

What you can now do in Power Query

- Append queries; names must match
- Combine a folder of files
- Split columns; fix error cells
- Merge; check rows, groups and totals

SAVE YOUR CHECKPOINT

D17-checkpoint.pbix, or browser report D17-checkpoint. Keep the data folder for refresh.